UNIVERSAL HOMEOMORPHISMS OF ALGEBRAIC SPACES TESTED BY BASE CHANGE ON TOPOLOGICAL SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a two-direction argument; the second direction names the three constituent properties of a universal homeomorphism and closes with universal closedness.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Universal homeomorphisms in the representable sense. Section section-representable, used at proof lines 11102-11108 and 11114-11120. Fixes the property being characterized.
- A universal homeomorphism of schemes is integral, universally injective and surjective. Morphisms, Lemma morphisms-lemma-universal-homeomorphism, used at proof lines 11120-11124. Supplies the three properties of the base changes to schemes.
- The general principle translating representable properties into scheme properties. Spaces, Lemma spaces-lemma-representable-transformations-property-implication, used at proof lines 11121-11123. Transfers the three properties back to f.
- An integral morphism is universally closed. Lemma lemma-integral-universally-closed, used at proof lines 11125-11127. Gives universal closedness, which combined with injectivity and surjectivity gives the homeomorphism statement.

Source reference. Original lemma 0415: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5280–5280. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04DF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5725–5725. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 03HA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4559–4559. Navigation pointer into the Stacks

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02YO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3883–3883. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a representable morphism of algebraic spaces over S. Then f is a universal homeomorphism (in the sense of Section 03HA) if and only if for every morphism of algebraic spaces $Z \to Y$ the base change map $Z \times_Y X \to Z$ induces a homeomorphism $|Z \times_Y X| \to |Z|$.*

Proof. If for every morphism of algebraic spaces $Z \to Y$ the base change map $Z \times_Y X \to Z$ induces a homeomorphism $|Z \times_Y X| \to |Z|$, then the same is true whenever Z is a scheme, which formally implies that f is a universal homeomorphism in the sense of Section 03HA. Conversely, if f is a universal homeomorphism in the sense of Section 03HA then $X \to Y$ is integral, universally injective and surjective (by Spaces, Lemma 02YO and Morphisms, Lemma 04DF). Hence f is universally closed, see Lemma 0415 and universally injective and (universally) surjective, i.e., f is a universal homeomorphism. □